

if conditions on p 117 are met

① then the integral operator

$$T(y)(x) = b + \int_a^x g(x, y(x)) dx$$

is a contraction on $X = C(I, J)$

$$I = [a - \delta, a + \delta] \quad J = [b - \epsilon, b + \epsilon]$$

② then the ODE

$$y' = g(x, y) \quad y(a) = b \quad \text{has}$$

unique solution

page 117 conditions are hard to apply
so we develop easier version.

Summary of the 2 sufficient conditions:

① $\delta \cdot \sup_{(x,y) \in I \times J} |g(x,y)| \leq \epsilon$

② $\delta \cdot \sup_{x \in I} |g(x, y_1(x)) - g(x, y_2(x))| < c \cdot \sup_{x \in I} |y_1(x) - y_2(x)|$

$I = [a-\delta, a+\delta]$

$J = [b-\epsilon, b+\epsilon]$

Question: Do these conditions hold? $c < 1$

① $\Rightarrow T(X) \subseteq X \quad y \in C[I, J] \Rightarrow T(y) \in C[I, J]$

~~$y \in C[I, J]$~~ $T(y)(x) = \int_a^x g(t, y(t)) dt + b.$

FIC $\Rightarrow$ $T(y)$ -continuous

$T(y)(x) \in J$?
if $x \in I$

$|T(y)(x) - b| < \epsilon$

$|T(y) - b| = \left| \int_a^x g(t, y(t)) dt \right| \leq \underbrace{|x-a|}_{< \delta} \cdot \max |g|$

$\left| \int_a^v h(t) dt \right| \leq |v-u| \cdot \max_{t \in [a,v]} |h(t)|$
Ineq ①

$|T(y) - b| \leq \delta \cdot \max_{x,y \in I \times J} |g(x,y)| \leq \epsilon \Rightarrow$

$\Rightarrow T(y)(x) \in J.$

Definition

Let $X \subseteq \mathbb{R}$. A function $f : X \rightarrow \mathbb{R}$ is called Lipschitz continuous if there is a $K > 0$ such that

$$|f(x) - f(y)| \leq K|x - y|, \forall x, y \in X.$$

The number K is then called a Lipschitz constant for f .

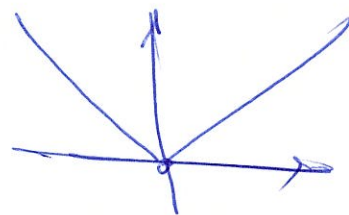
~~$f \in C(\mathbb{R})$~~ $f \in C^1(\mathbb{R})$ bounded derivative
 then its Lipschitz
 by Mean Value Theorem $f' \in C(\mathbb{R})$
 $f(x) - f(y) = f'(c)(x - y) \quad x < c < y$

$$|f(x) - f(y)| \leq K|x - y| \quad K = \sup_{c \in \mathbb{R}} |f'(c)| < \infty$$

$$f(x) = x^2 \in C^1(\mathbb{R}) \quad f'(x) = 2x \quad \text{not bounded}$$

$$f(x) = |x| \quad \text{not differentiable}$$

$$|x| \text{ is Lipschitz}$$



$$f(x) = x^{1/2} \quad \text{not Lipschitz}$$

$$|f(x) - f(0)| \leq K|x - 0| \Leftrightarrow$$

$$\Leftrightarrow |x^{1/2}| \leq K \cdot |x| \Leftrightarrow |x^{-1/2}| \leq K \quad \forall x$$

Exercise: Suppose f is continuous on a closed interval $[a, b]$, differentiable on (a, b) , and f' is bounded on (a, b) . Then f is Lipschitz continuous on $[a, b]$.

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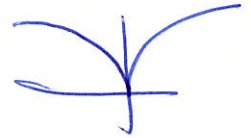
Exercise: Give an example of a continuous function on a closed interval which is not Lipschitz continuous.

bounded

$$f(x) = x^{1/2}$$

$\mathbb{R}$

$$[-1, 1]$$



$$f(x) = x^{1/3}$$

$$[-1, 1]$$

see page 118

Definition

If $X \subseteq \mathbb{R}^2$, then $f : X \rightarrow \mathbb{R}$ is Lipschitz continuous in the second variable if there is a $K > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq K \cdot |y_1 - y_2|, \quad \forall (x, y_1), (x, y_2) \in X.$$

② test in P-L-T reduces to
or $\nabla \circ \text{MVT}$ added

$$K = \sup_{x, y \in X} \left| \frac{\partial f}{\partial y} \right|$$

We can now simplify the two sufficient conditions above as follows: For the function g on the rectangle $I \times J$, let M be an upper bound, and let K be a y -Lipschitz constant. Then

① Condition (1) above holds if $\delta < \frac{\epsilon}{M}$

② Condition (2) above holds if $\delta < \frac{\epsilon}{K}$

①+② here are
the simplest conditions
suitable for P-L-T.

$$M = \sup |g(x, y)|$$

$$K = \sup \left| \frac{\partial g}{\partial y}(x, y) \right|$$

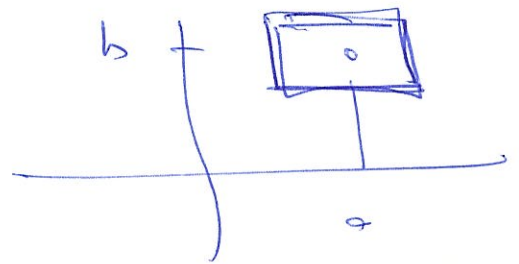
Theorem (Picard-Lindelöf Theorem)

Let g be a continuous function on a neighborhood of $(a, b) \in \mathbb{R}^2$ which is Lipschitz continuous in the second variable. Then there is an interval around a on which the differential equation

$$y' = g(x, y), \quad y(a) = b$$

has a unique solution.

$\frac{\partial g}{\partial y}$ continuous & bounded



Example: Consider the initial value problem

$$y' = \cos(xy), \quad \underline{y(0) = 0}.$$

Does this have a unique solution on a neighborhood of 0?

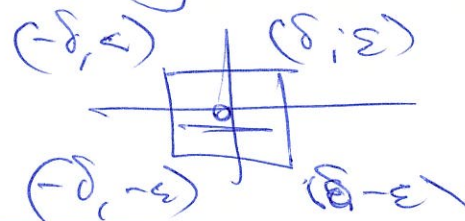
$$g(x, y) = \cos(xy)$$

$$\frac{\partial g}{\partial y} = -x \sin(xy)$$

$$\|g\|_{\infty} \leq 1 = M$$

$$|x| \leq 1$$

$$\left\| \frac{\partial g}{\partial y} \right\|_{\infty} \leq 1 = K$$



Example: Consider the initial value problem

$$y' = y^{\frac{1}{3}}, \quad y(0) = 0. \quad \leftarrow$$

Does this have a unique solution on a neighborhood of 0?

separable ODE

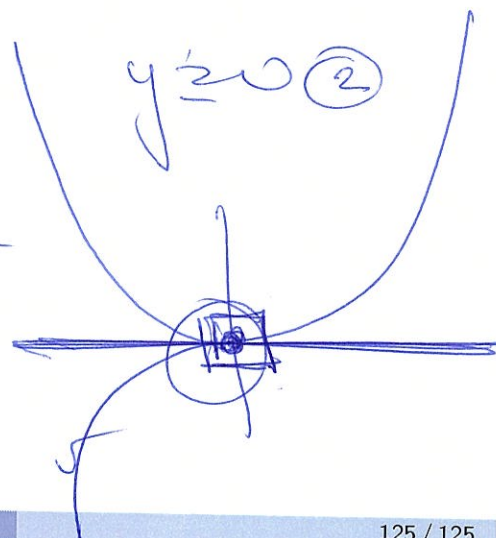
$$\frac{dy}{y^{\frac{1}{3}}} = dx$$

$$\frac{2}{3} y^{\frac{2}{3}} = x + C$$

$$\textcircled{1} \quad y = \left(\frac{3}{2} x + C \right)^{\frac{3}{2}}$$

$C=0$

$$g(x, y) = y^{\frac{1}{3}}$$



$$\frac{\partial g}{\partial y} = \frac{1}{3} y^{-\frac{2}{3}}$$